# FRA Diagnostic Report

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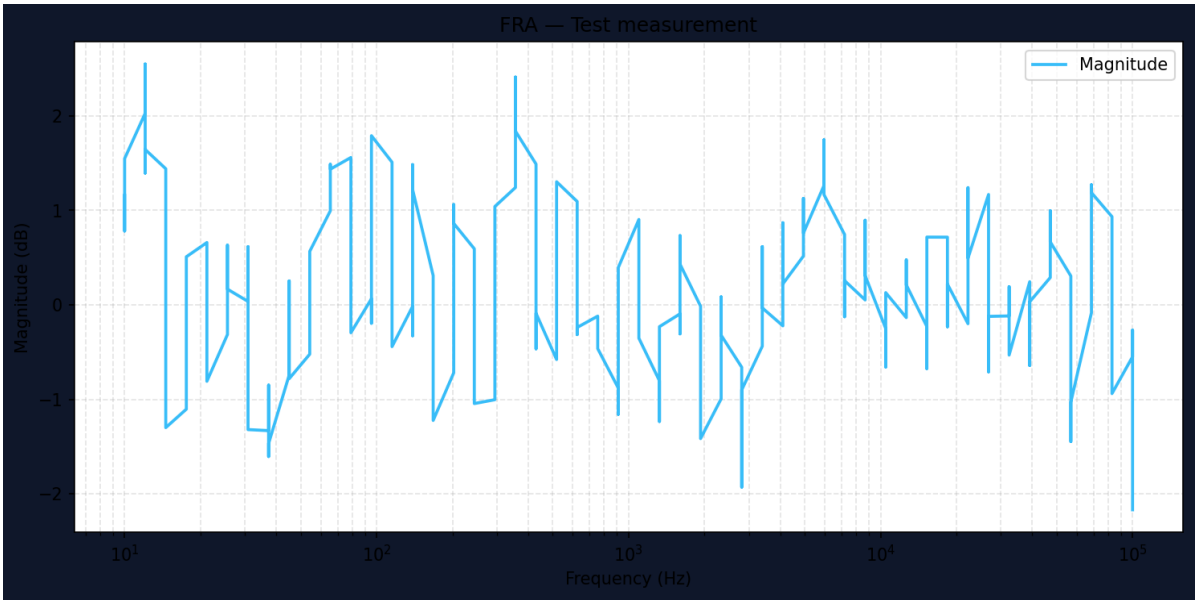
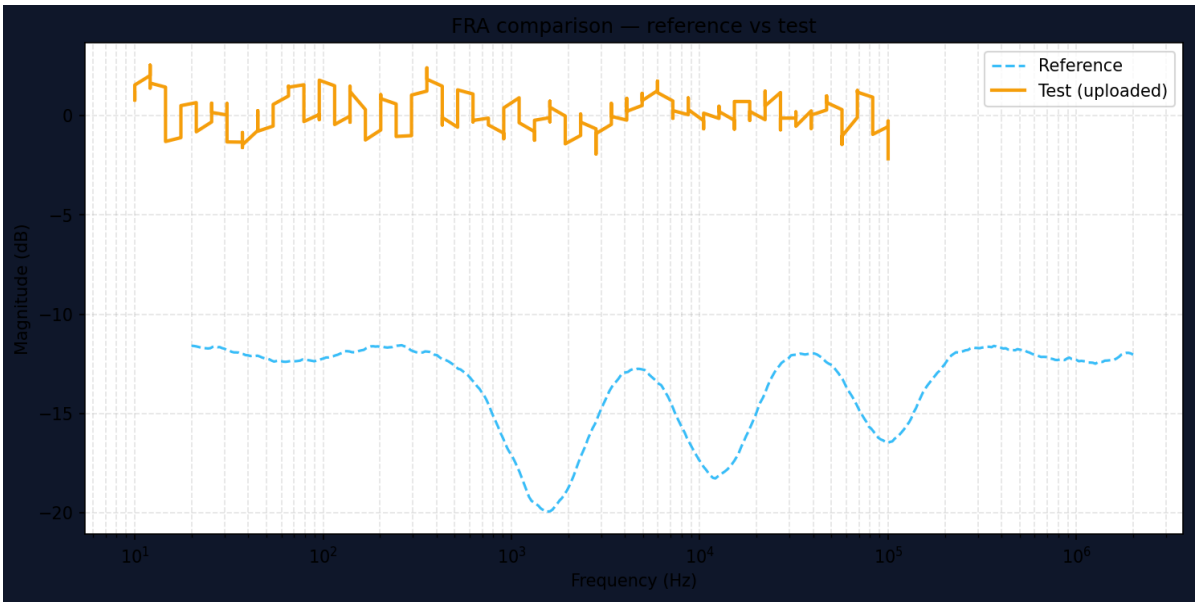
| Field | Value |
|---|---|
| Fault assessment | Winding Deformation |
| Confidence (0–100) | 44.9 |
| Severity | High |
| Anomaly score (0–100) | 100.0 |
| Correlation (vs ref) | 0.1694 |
| Max deviation | 20.28 dB |

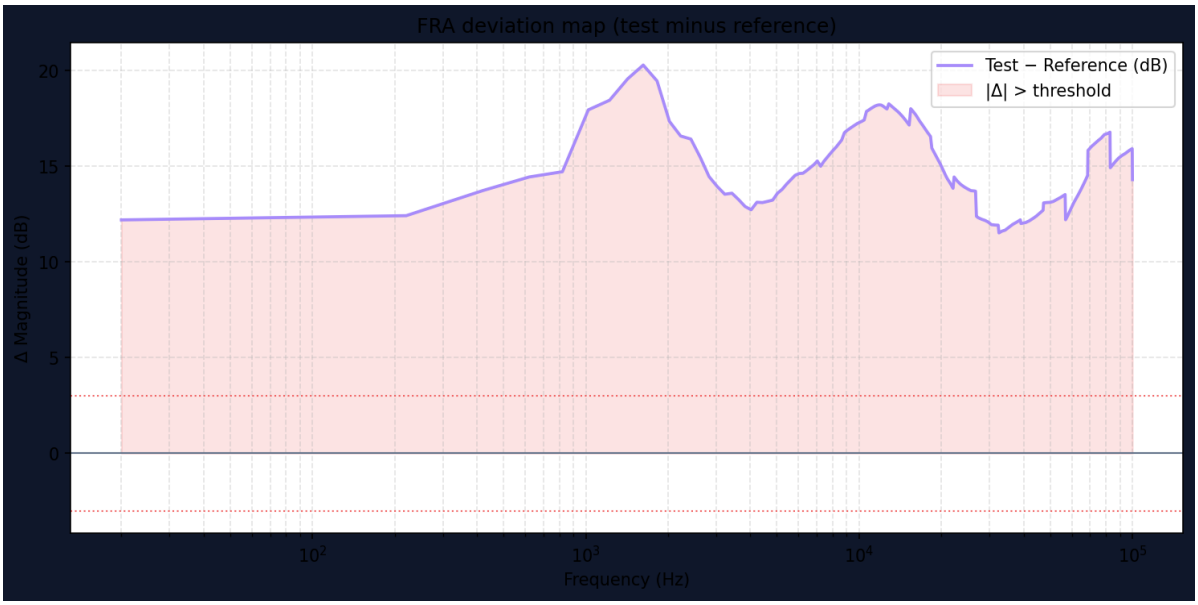
### *Single FRA (test)*




### *Comparison*




### *Difference map*




### *Recommendation*

Significant deviation detected in mid-frequency range (10kHz–100kHz) suggesting axial winding displacement. Schedule internal inspection within 30 days. Reduce loading to 80% capacity until inspection is completed. Expert notes: Low-frequency energy deviation → possible core or clamping issue; Broadband magnitude deviation vs reference

### *Explanation*

ML classifier: Winding Deformation (73% estimated class confidence). Statistical correlation vs reference: 0.225. IsolationForest anomaly score (0–100, higher=worse): 100.0. Rules: Low-frequency energy deviation → possible core or clamping issue | Broadband magnitude deviation vs reference